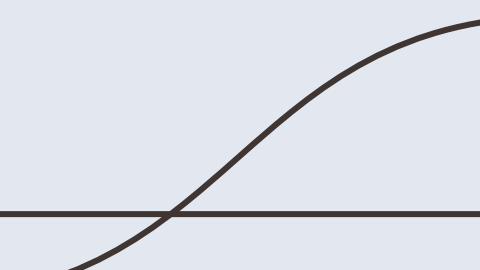


The Impact of Academic Pressure on Anxiety and Sleep Among AP Students

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“To what extent does perceived academic pressure predict anxiety and sleep deprivation among AP students at competitive high schools?”

Hypotheses

- For the anxiety test, our null hypothesis (H_0) will be that there is no relationship between academic pressure and anxiety.
 - Our alternative hypothesis (H_a) will be that academic pressure is associated with higher anxiety.
- For the sleep deprivation test, our null hypothesis (H_0) will be that academic pressure is not related to sleep deprivation.
- Our alternative hypothesis for the sleep deprivation test (H_a) is that academic pressure will lead to less sleep and greater sleep deprivation.

Method- Quantitative Correlational Study

- Anonymous online survey distributed to AP students
 - Random sampling procedure:
 - 5 AP Classes randomly selected.
 - Random number generator used
 - 10 Students randomly selected from each class and given survey.
- Participants:
- n = 50 AP students
 - All currently enrolled in at least one AP course

Method

Survey Structure

- 5-point Likert scale (Strongly Disagree → Strongly Agree)

Measured:

- Perceived academic pressure
- Anxiety levels
- Sleep habits

Composite scores created for:

- Pressure
- Anxiety
- Sleep deprivation

Data Analysis

Calculated:

- Mean & standard deviation
- Correlation (r)
- Regression line

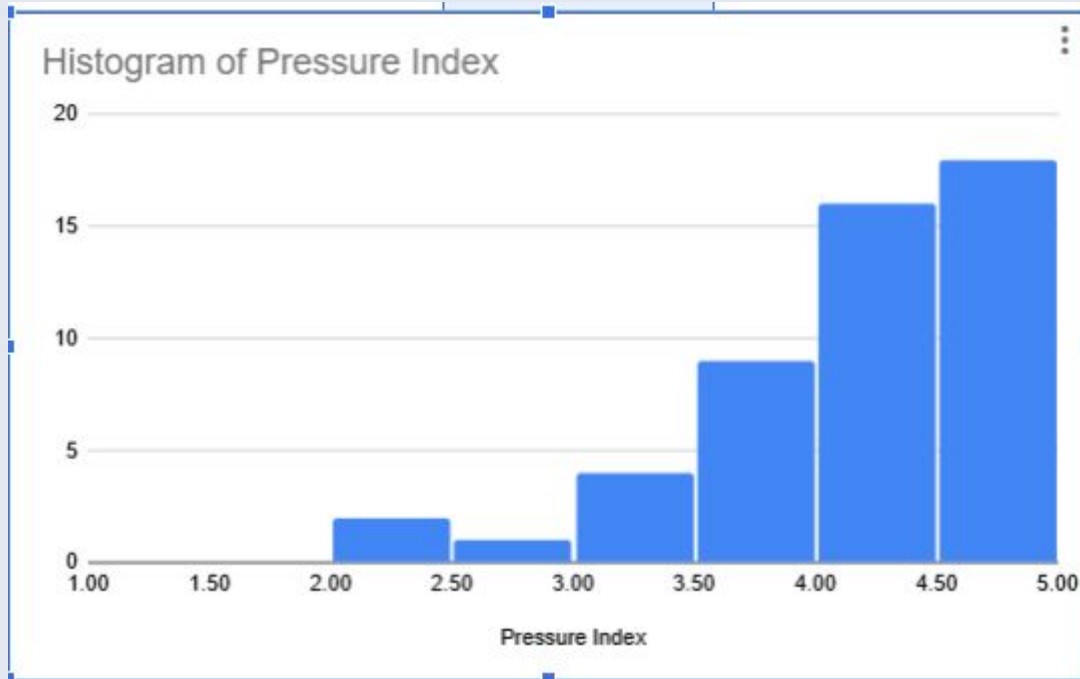
Tested if academic pressure significantly predicts:

- Anxiety
- Sleep deprivation

Example Survey Question

- I feel constant pressure to earn high grades in my AP courses.
- I feel stressed about maintaining a competitive GPA.
- I have difficulty relaxing because of academic responsibilities.
- On average, I get fewer than 7 hours of sleep on school nights.

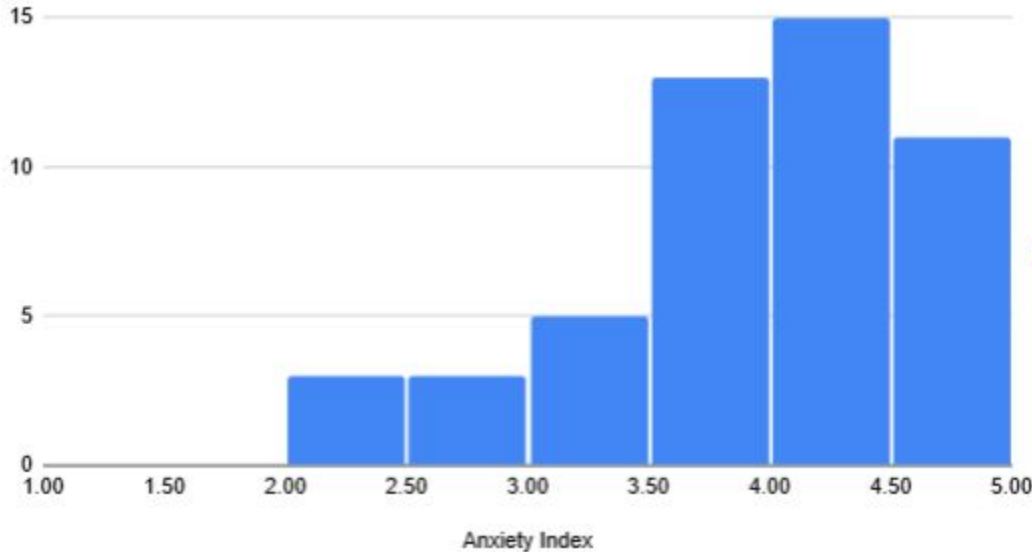
Descriptive Statistics: Pressure



- Mean: 3.82
- SD: 0.7610411772
- Min: 2.33, Max: 5.00
- Median: 4
- Q1: 3.5, Q3: 4.5
- Distribution: left-skewed,
highly concentrated
around 4–4.5

Descriptive Statistics : Anxiety

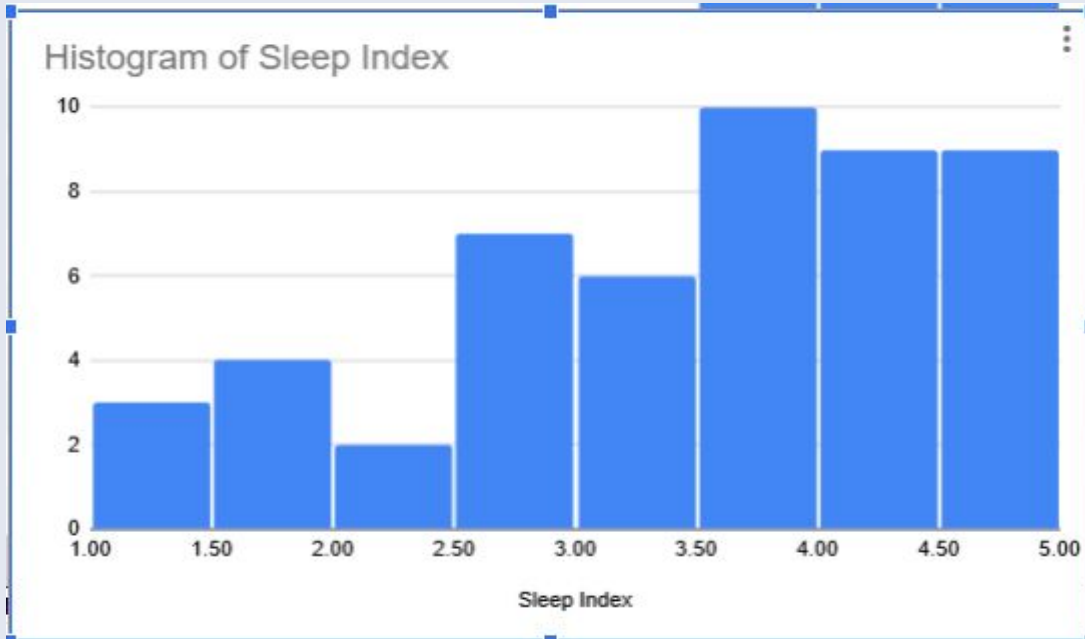
Histogram of Anxiety Index



- Mean: 3.49
- SD: 1.085573299
- Min: 2.00, Max: 5.00
- Median: 4.00
- Q1: 3.50, Q3: 4.5
- Distribution: left-skewed with more variability

Descriptive Statistics

Sleep

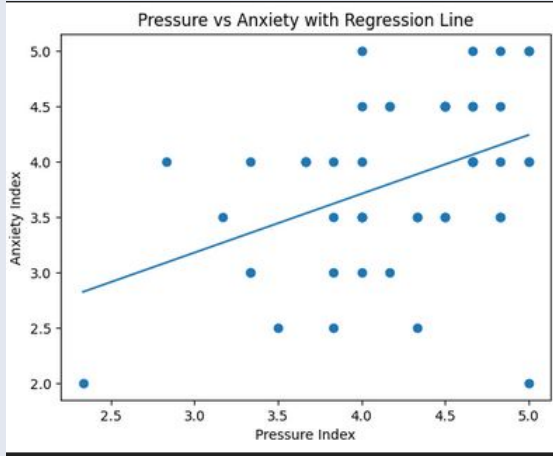


- Mean: 3.49 SD: 1.08
- Min: 1.00, Max: 5.00
- Median: 3.5
- Q1: 3.00, Q3: 4.55
- Distribution: most spread out, full range 1-5

Proportion Analysis

- High Anxiety (Index ≥ 4): **29/50 = 0.58** Insufficient Sleep (Index ≥ 4): **23/50 = 0.46**
- Indicates widespread stress and sleep disruption among AP students

Pressure and Anxiety



Regression Equation:

$$\hat{Y} = 1.585 + 0.531X$$

- Slope interpretation: Each 1-unit increase in pressure predicts a 0.531-unit increase in anxiety
- intercept not meaningful in context
- Pressure explains 18.0% of anxiety variation

Conditions Check

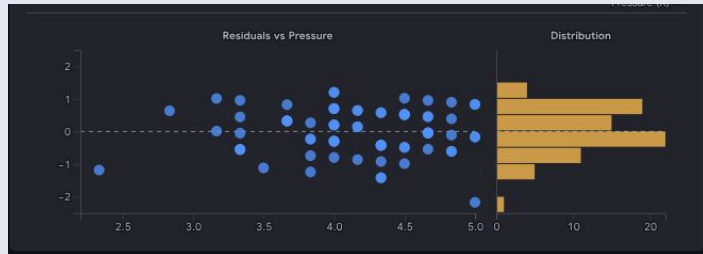
- Linearity: scatterplot shows roughly linear pattern
 - Independence: sample size < 10% of population
 - Normality: residuals show no skew or outliers
- Equal variance: residual plot show consistency spread

Hypothesis Test

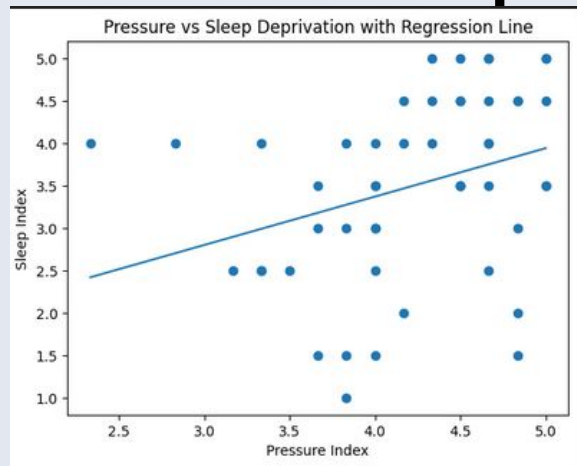
$$H_0: B = 0, H_a: B > 0$$

- $t = 3.25, df = 48$
- $p = 0.0044$
- Conclusion: $p < 0.05 \rightarrow$ reject .

Academic pressure is a significant positive predictor of anxiety



Pressure and Sleep



Regression Equation

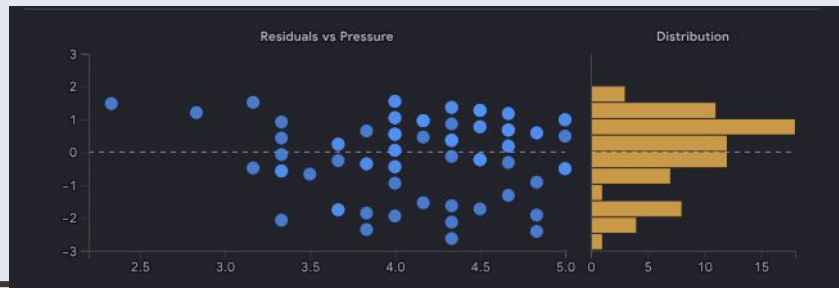
- $\hat{Y} = 1.089 + 0.570X$
- Slope interpretation: Each 1-unit increase in pressure predicts a **0.57-unit** increase in sleep-deprivation score
- Intercept not meaningful
- $R^2 = 0.10227$: Pressure explains **10.2%** of sleep-deprivation variation

Conditions Check

- Linearity: scatterplot shows linear trend
- Independence: sample size < 10% of population
- Normality: residuals show no outliers
- Equal variance: residual plot shows consistent spread

Hypothesis Test

- $t = 2.34$, $df = 48$
- $p = 0.015954$
- **Conclusion:** $p < 0.05 \rightarrow$ reject H_0 .
Academic pressure is a **significant positive predictor** of sleep deprivation



Final Conclusion

- Although the coefficient of determination was relatively small, this is typical for behavioral research. Cohen (1988) notes that correlations around $r = 0.30$ are considered moderate effects in psychological studies, and many educational and stress-related studies report similar magnitudes.
- Academic pressure is high among AP students Pressure significantly predicts both anxiety and sleep deprivation
- Both relationships are positive and statistically significant
- Schools may benefit from stress-management and workload-balancing strategies

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